

Simone Carletti

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About Me

I am a M.Sc. student in **Computer Engineering** specializing in **Robotics and Autonomous Systems**. I am currently completing my master's thesis at EPFL, where I am developing an adaptive flight control system for a **morphing-wing drone**. This work has given me hands-on experience with Reinforcement Learning, Control, and UAV dynamics. I have a long-standing passion for **hardware**, having worked extensively with embedded platforms such as Arduino, RPi, ESP8266, etc. Beyond my academic work, my **professional experience** as a lead developer has allowed me to build strong software engineering skills, focusing on clean and maintainable code. I am proficient in **C++** and **Python**, with extensive experience with **Linux** environments, utilizing **Git**, and **debugging** complex multi-threaded systems.

Education

EPFL - LIS, Master's Thesis - Title: $\mathcal{L}_1$-Adaptive augmentation for robust morphing-wing drone flight control - Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS2, Git, Linux, Unity	<i>Sept 2025 – Present</i>
DTU, M.Sc. Autonomous Systems Erasmus+ exchange - Converted GPA: 30/30	<i>Sept 2024 – Jul 2025</i>
Politecnico di Torino, M.Sc. Computer Engineering Curriculum in Automation and Intelligent Cyber-Physical Systems - Final GPA: 29.48/30	<i>Sept 2023 – Present</i>
Università degli Studi di Firenze, B.Sc. Computer Science and Engineering Graduated with 110/110 cum laude (with honors)	<i>Sept 2020 – Sept 2023</i>

Selected Projects

Multiplayer videogame on embedded hardware (GitHub 📄) - Developed bare-metal C firmware on ARM Cortex-M3 (LandTiger) using Keil μVision - Designed and implemented a custom real-time communication protocol over CAN bus to enable multiplayer support on separate boards. - Ensured deterministic behavior under strict timing and memory constraints.	<i>Jan 2024 – Jun 2024</i>
Digital Control - Autonomous Robotics (GitHub 📄) - Designed, simulated, and tuned digital control systems using MATLAB and Simulink to ensure robust mathematical performance before hardware deployment. - Implemented the finalized control algorithms in C/C++ on an embedded microcontroller, successfully bridging the gap between theoretical simulation and real-time execution.	<i>Oct 2024 – Dec 2024</i>
Localization with EKF in ROS2 Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry	<i>Nov 2023 – Dec 2023</i>
Autonomous Navigation & Task Execution - DTU RoboCup 📄 Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform	<i>Feb 2025 – May 2025</i>

Experience

Robotics Engineer, DIANA student team Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration	<i>Sept 2023 – Jul 2024 Turin, Italy</i>
Full-Stack Developer, Genomedics S.r.l - Lead developer of three key projects and for the company's main project (KPlatform) - Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux	<i>Sept 2020 – Present Florence, Italy</i>

Skills & Activities

- Languages:** Fluent in English, Native Italian speaker, Basic Spanish skills
- Programming:** C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- Tools:** PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia Jetson/Orin, Guitar, Photography, Networking, Zigbee